Subject Description Form

Subject Code	ME5301
Subject Title	Embodied Robot Intelligence
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil.
Objectives	To equip students with the fundamental knowledge and skills required to design, develop, and integrate intelligent robotic systems that leverage artificial intelligence techniques for enhanced perception, autonomy, and human-robot interaction.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Explain the fundamental components and architecture of embodied robot intelligence. b) Apply AI algorithms and embodied learning techniques to enable intelligent decision-making in mechanical robots. c) Integrate sensor data and implement computer vision algorithms for enhanced perception in robots. d) Acquire proficiency in using software tools like Python, OpenCV, and ROS for robotics development. e) Develop intelligent robotic systems for real-world applications using the principles of computational robotics and AI.
Subject Synopsis/ Indicative Syllabus	**Robot Intelligence Models.** Fundamentals of embodied intelligence, deep robot learning, cognitive architectures for robotics, world models, policy representations and optimisation. **Intelligent Perception-Action Coupling.** Active robot vision, neural processing of visual/motor data, sensorimotor integration, adaptive robot behaviour, intelligent grasping and manipulation, robot learning from demonstrations, motion control via reinforcement learning. **Generative Robot Intelligence.** Overview of robot GenAI, generative design of robot morphologies, automated robot task specification and planning, grounding mechanisms and prompt engineering, synthetic data generation for robot training, generative creation of robot simulations, intuitive human-robot interactions. **Software Tools and Frameworks.** Python, OpenCV, common Foundation Models and fine-tuning tools, Robot Operating System (ROS). **Practical Work.** Validate embodied robot AI in simulation environments, group projects and practical cases of study.

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<td>Teaching/Learning Methodology</td>
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1. Lectures will be used to provide students with theoretical knowledge required for an advanced in-depth study of the field of embodied robot intelligence (Outcomes a to e).
2. Tutorials enhance analytical and problem-solving skills, with a focus on sensor integration, computer vision, and software tools (Outcomes a to e).
3. Hands-on projects allow students to develop algorithms for robots and apply computational principles to solve problems (Outcome a to e).

Teaching/Learning Methodology	Intended Subject Learning Outcomes to be assessed				
	a	b	c	d	e
1. Lecture	√	√	√	√	
2. Tutorial	√	√	√	√	√
3. Experiments/Project	√	√	√	√	√

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<tr>
<td>Assessment Methods in Alignment with Intended Learning Outcomes</td>
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Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
		a	b	c	d	e
1. Test	10%	√	√	√		√
2. Coursework	50%	√	√	√	√	√
3. Final Examination	40%	√	√	√		√
Total	100 %					

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Overall Assessment:

0.4 x End of Subject Examination + 0.6 x Continuous Assessment.

1. The continuous assessment aims at evaluating the progress of the students’ study, assisting them in self-monitoring the respective learning outcomes, and applying the knowledge learnt in practical situations.
2. The examination is used to assess the knowledge acquired by the students for understanding and analysing the problems critically and independently; as well as to determine the degree of achieving the subject learning outcomes.

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Student Study Effort Expected	Class contact:	
	▪ Lectures	33 Hrs.
	▪ Tutorials / Laboratory	6 Hrs.
	Other student study effort:	
	▪ Course work	30 Hrs.
	▪ Self-learning	36 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	1. How the Body Shapes the Way We Think, Rolf Pfeifer and Josh Bongard, MIT Press, latest edition. 2. Embodied Cognition, Lawrance Shapiro, Routledge, latest edition. 3. Introduction to AI Robotics, Robin Murphy, MIT Press Cambridge, MA, USA, latest edition. 4. S. Shalev-Shwartz and S. Ben-David. Understanding Machine Learning: From Theory to Algorithms, latest edition. 5. Goodfellow, Y. Bengio, and A. Courville, Deep Learning, MIT Press, latest edition.	
Develop Date	February 2025	